

Lecture 1: Introduction to Information Security

Eleftheria Makri | Security

04/02/2026



Universiteit
Leiden

Today's agenda

- Organizational details
 - *How to succeed in learning and passing the course*
- Topic today: **Introduction to Information Security**
 - What is security?
 - Security trade-offs
- Plan of the course

Organizational details

- **Lecturers:**

- Eleftheria Makri
- Francesco Bariatti

- **TA Team:**

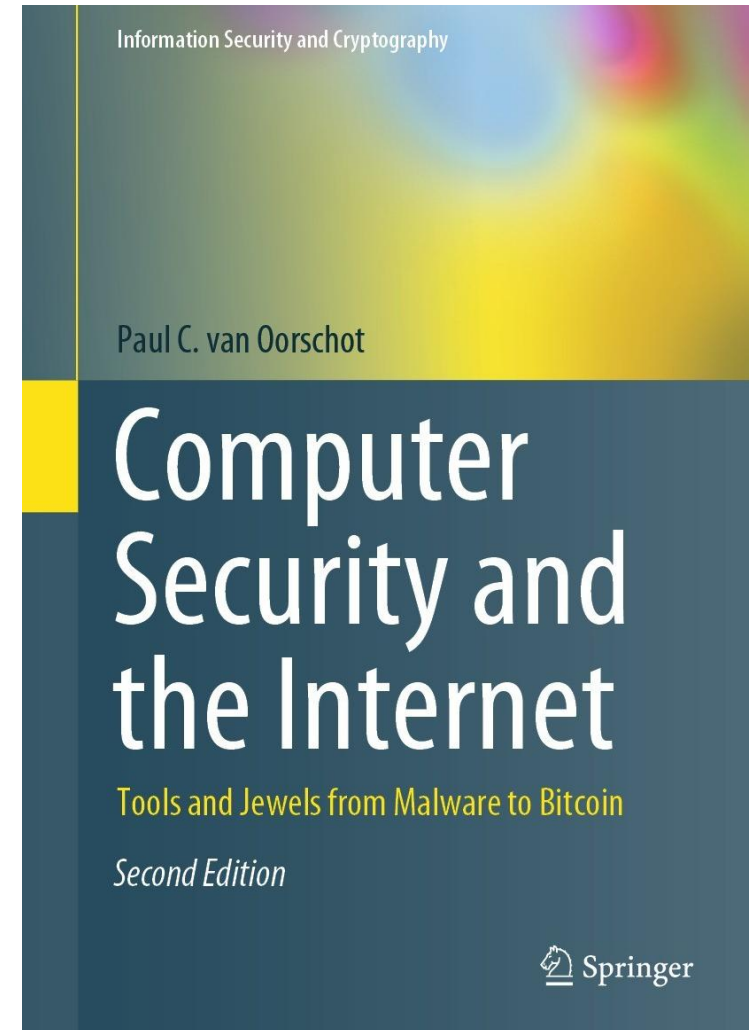
- Anna
- Cesar
- Daan
- Dheeraj
- Jayme
- Laith
- Nicolas
- Teun

Contact the lecturer <e.makri@liacs.leidenuniv.nl>
regarding course organization
(deadline extension requests, questions about lecture
material, readings, etc.)

Email to contact about assignment questions and issues:
<security2026@liacs.leidenuniv.nl>

Literature and Readings

- Recommended but not required textbook:
 - P. van Oorschot “*Computer Security and the Internet. Tools and Jewels*” Springer, 2021 [2020 is also fine]. PDF is available at the author’s webpage.
 - Textbook is intended as support material; not as required reading.
- Other additional readings will be provided per lecture.
- Lecture slides and some material will be provided on Brightspace.



Course Structure & Success Criteria

- Lectures (x14) - learning theory of security
 - In-person; no video recordings or online streaming
- Practicals (x13) - working on practical assignments
 - In the Gorlaeus Building on Wednesdays 11.00 – 12.45 (in principle; for exceptions see timetable)
- Average of 3 assignments —> **Assignment grade**
- Written exam —> **Exam grade**
- **Final grade** = 55% Exam grade + 45% Assignment grade
 - Both the final exam grade and the average grade of the 3 assignments must be at least 5.5
 - Assignments are re-takeable towards the end of the semester, but we **STRONGLY** advice avoiding them (by passing the first chance), as the retakes are significantly more challenging than the first batch of exercises.
 - Exam resit is allowed

Assignments

- **Release dates and deadlines:**
 - Ass. 1 - February 18 – March 12: Bruteforcing
 - Ass. 2 - March 11 – April 16: Scenario-based vulnerability tracing (option 1)
 - Ass. 2 - March 11 – April 16: Application security (option 2)
 - Ass. 3 - April 15 – May 14: Web security
 - Retake - May 14 – May 28: Retake assignments (any)
- Use the practicals to work on assignments. TAs are there to help you.
- For all assignments submit a report and all requested supplementary material via ANS. You may work in **groups of 2 students** (recommended), or **individually** on each of the assignments. You may change your group for each assignment.
- **Deadline extensions** will be granted by the lecturer strictly individually and in special circumstances, and only if requested at least 2 days prior to the deadline.
- **Late submissions will be penalized as follows:**
 - Minus 1 point (on the 10-point scale) for each 24 hours of delay, starting from 5 minutes. Submitting 10 min late: -1 point; 25 hours late: -2 points, etc.

Assignments

Bruteforcing (Feb 18 – Mar 12)

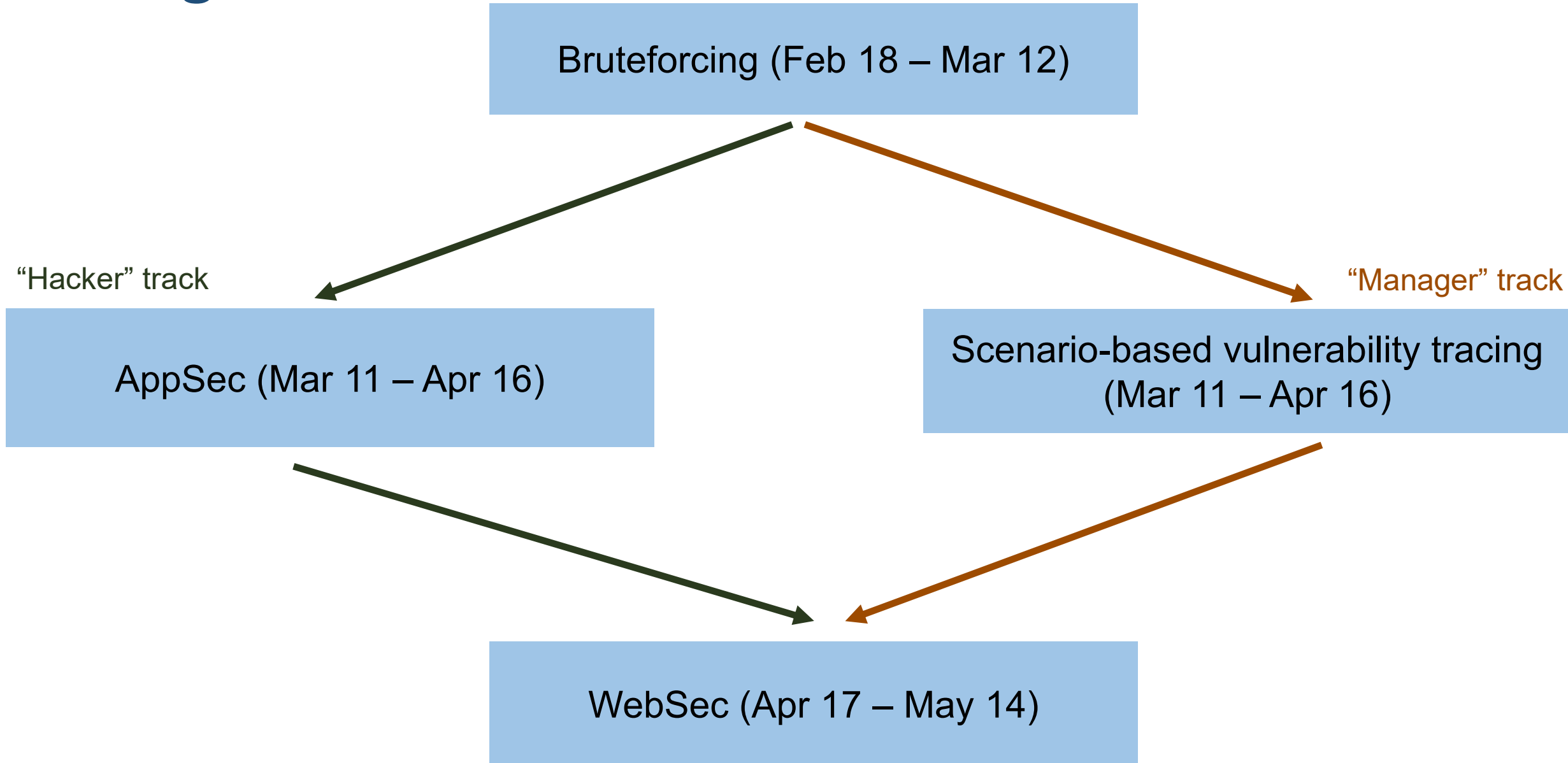
“Hacker” track

AppSec (Mar 11 – Apr 16)

“Manager” track

Scenario-based vulnerability tracing
(Mar 11 – Apr 16)

WebSec (Apr 17 – May 14)



- Your assignments will focus on hard skills and soft skills.

- Reporting is an important task in any cyber security career path.
- Security tasks are not well-defined by nature (hence creativeness).

<https://www.varonis.com/blog/working-in-cybersecurity/>



Plagiarism

- Assignments are in **individual** or **groups of 2**.
- **Do** discuss and help each other on the course content.
- **Do not** share (partial) solutions to assignments outside your group.
- **Acknowledge** all your sources.
- Copying (parts of) solutions, from another group or from any other source is plagiarism.
 - We will report all plagiarism cases to the Board of Examiners.
 - Both who copied and who shared their solution will face consequences.

Why Learning Security?

Information Security – 2023 Breaches



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More

Shell Recharge security lapse exposed EV drivers' data

Zack Whittaker @zackwhittaker / 9:30 AM GMT+2 • June 9, 2023

Comment



Image Credits: Chris Ratcliffe / Bloomberg / Getty Images



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Information Security – 2023 Breaches

RTLnieuws

☀ 27°

🚗 5 km

🚆 1 OV



Ransomware-aanval

Paspoorten van dokters op straat na hack bij oudereninstelling Gelderland



Door Daniël Verlaan

7 maart 2023 12:42 • Aangepast 7 maart 2023 15:04

Beeld © ANP



Beeld ter illustratie.



Cybercriminelen hebben de paspoorten van dokters, verpleegkundigen en fysiotherapeuten gelekt die werkzaam zijn of waren bij Attent Zorg en Behandeling, een instelling voor ouderenzorg in Gelderland. De organisatie werd in februari slachtoffer van een ransomware-aanval.



Net binnen

- 14:14 Roeiers Holland Acht met tweede plaats naar WK-finale
- 14:10 Veel ophef na uitlatingen Van Gaal over WK-duel met Argentinië
- 14:10 Tienduizenden reizigers dupe van failliete Holland Norway Lines
- 14:05 Willem II ontslaat trainer Robbemonnd
- 14:00 Reijnders blij dat AZ na vertrek dragende spelers Europa in gaat

Meer nieuws

STAP DAN
OVER
NAAR

Information Security – 2023 Breaches

MOVEit, the biggest hack of the year, by the numbers

At least 60 million individuals affected, though the true number is far higher

Carly Page @carlypage_ / 5:45 PM GMT+2 • August 25, 2023

Comment



Image Credits: Frank Ramspott / Getty Images

The mass-exploitation of MOVEit Transfer software has rapidly cemented itself as the largest hack of the year so far. While the full impact of the attack will likely remain untold for months to come, there are now more than 1,000 known victims of the MOVEit breach, according to cybersecurity company Emsisoft.

Information Security – 2023 Breaches

RTLnieuws

☀️ 27° 🚗 4

Ernstig lek bij Kadaster: miljoenen woonadressen door iedereen op te zoeken



Door Daniël Verlaan

28 augustus 2023 14:00 • Aangepast 29 augustus 2023 10:30



Iedereen die een koophuis heeft is door het lek geraakt.



Het was lange tijd kinderlijk eenvoudig om toegang te krijgen tot het besloten deel van het Kadaster. Normaal gesproken hebben bijvoorbeeld notarissen en gerechtsdeurwaarders toegang tot informatie zoals volledige namen en woonadressen, maar door een lek kon iedereen erbij. Levensgevaarlijk, zeggen experts.



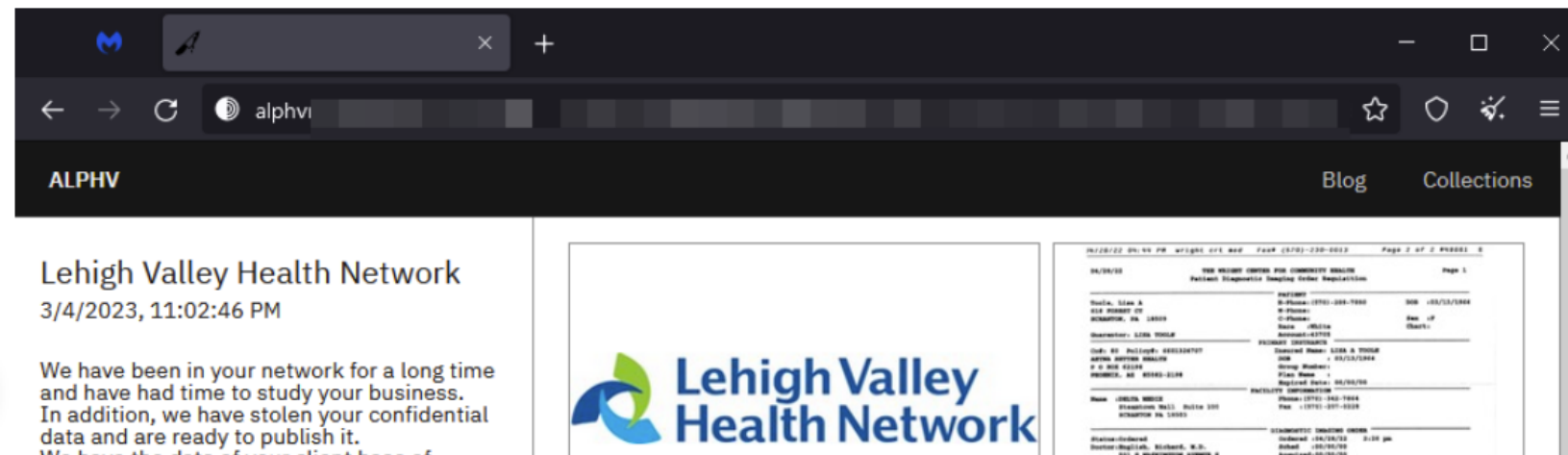
Information Security – 2023 Breaches

Breast cancer photos published by ransomware gang

Posted: March 13, 2023 by [Jovi Umawing](#)

The Russia-linked ALPHV ransomware group, also known as [BlackCat](#), has posted sensitive clinical photos of breast cancer patients—calling them "nude photos"—to extort money from the Lehigh Valley Health Network (LVHN).

This has triggered a chorus of accusations from the cybersecurity community, with some labeling the group as "[barbarians](#)" and others saying the group is "[exploiting and sexualizing breast cancer](#)".



Major Security Breaches of 2025

Breach	Organization	Records Affected	Data Compromised	Cause of Breach
1. Google, Apple, Facebook	Various	16 billion credentials	Username, password for major platforms	Compilation of stolen credentials through malware
2. Ahold Delhaize	Retail	Sensitive personal, financial info	Names, IDs, bank account info	Ransomware by INC Ransom group
3. Marks & Spencer	Retail	Not specified	Active Directory data, financial records	Ransomware via social engineering
4. Coinbase	Financial	69,461 users	Names, contact details, partial Social Security	Insider threats from support contractors
5. TransUnion	Finance	Not specified	Personal information	Cyber attack linked to third-party vulnerabilities
6. Red Hat Consulting	Technology and Government	570 GB of data	VPN settings, API keys, security configurations	Supply chain compromise by Crimson Collective

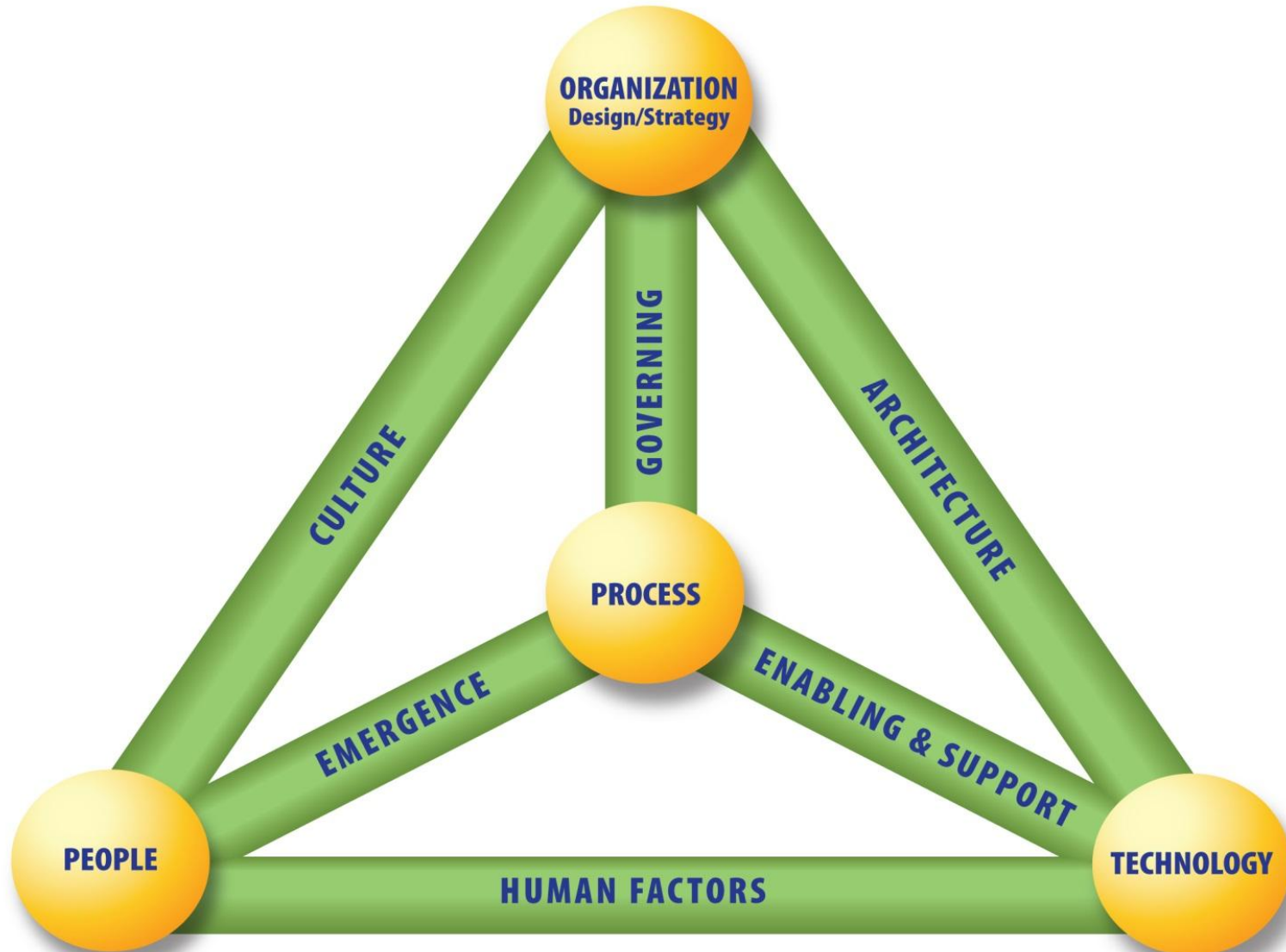
Computer Security (COMPUSEC)



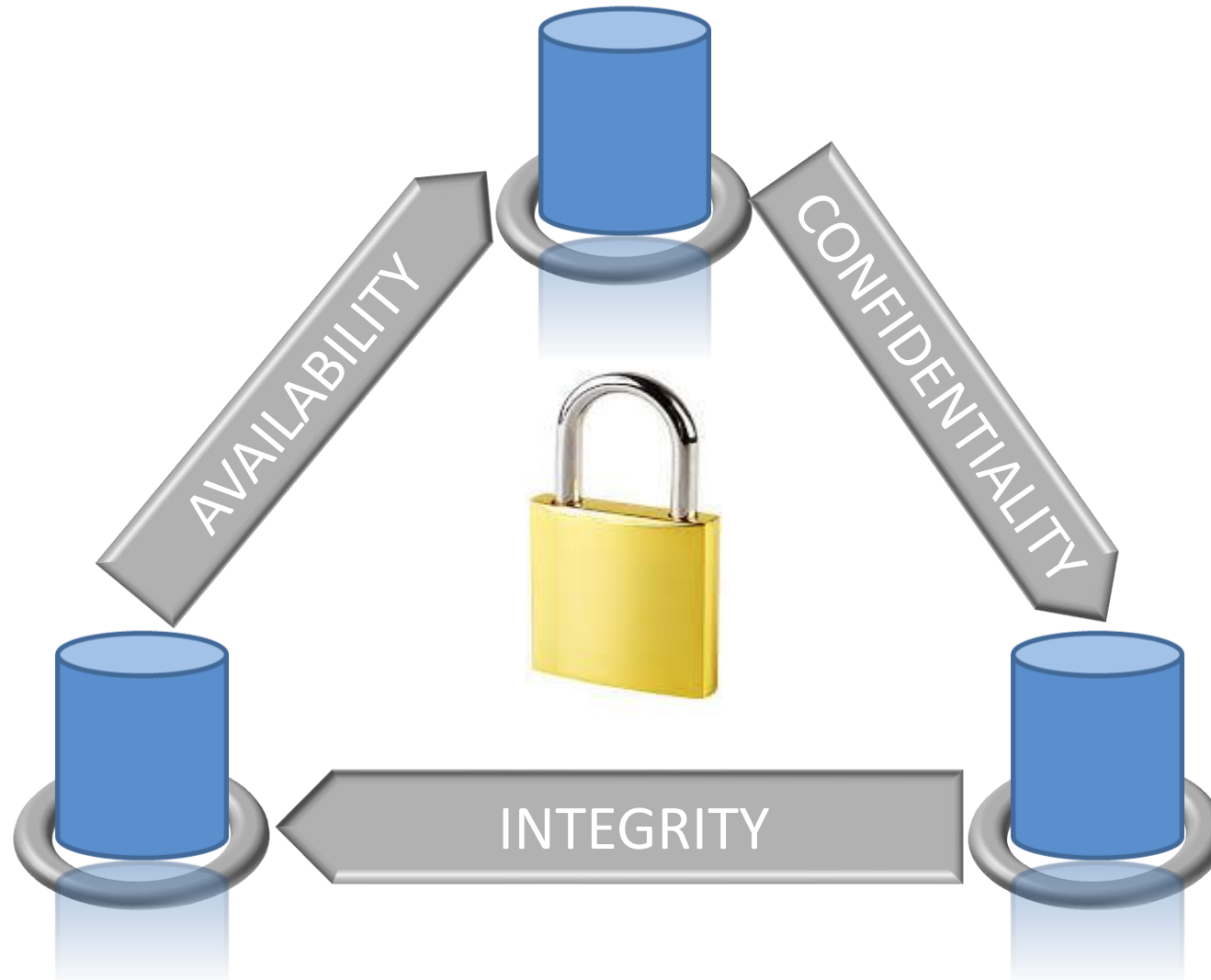
Information Systems Security (INFOSEC)



Information Security



Security Principles



Additional Security Properties

(including the AAAA properties)

Authenticity



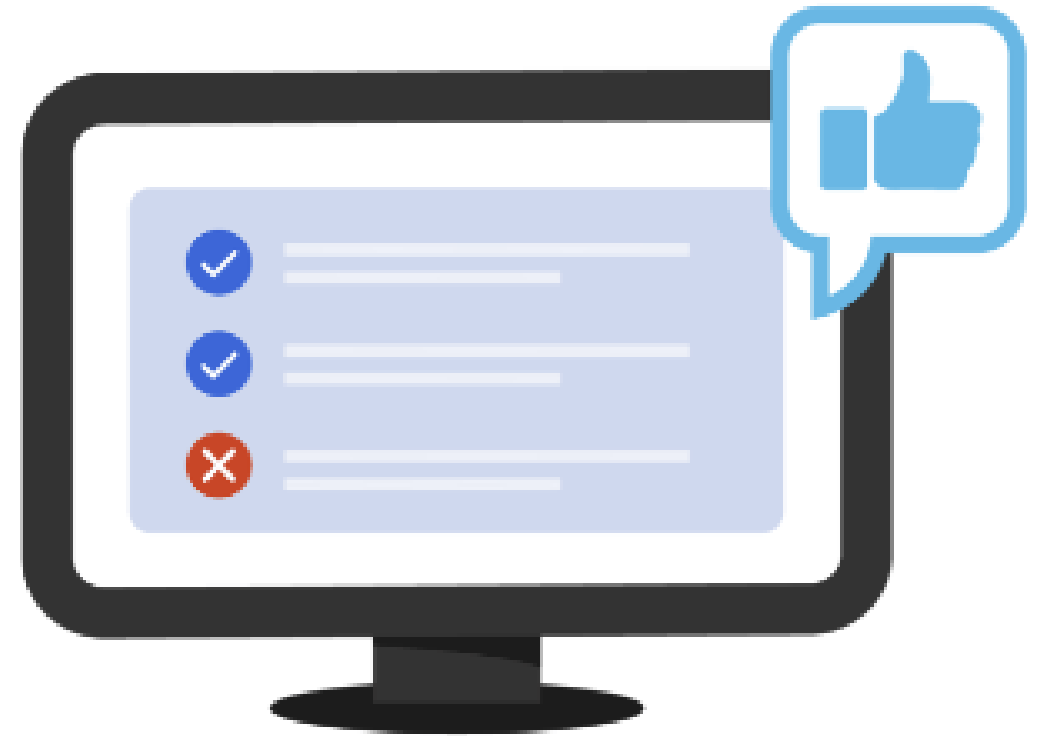
Authorization

Authentication



Confirms users
are who they say they are.

Authorization



Gives users permission
to access a resource.

Accuracy



Accountability

✕

—

□

syslog

updated today 07:12:23 PM

⚙

🔍

▶ auth.log

▶ dpkg.log

▶ mail.log

▶ **syslog**

▶ Xorg.0.log

Jan 11 19:12:02 virtualbox-ivan-ubuntu-16

kernel: [112.397183] ISO 9660 Extensions: RRIP_1991A

Jan 11 19:12:02 virtualbox-ivan-ubuntu-16

udisksd[3686]: Mounted /dev/sr0 at /media/ivan/VBox_GAs_5.2.2 on b

Jan 11 19:12:02 virtualbox-ivan-ubuntu-16

dbus[742]: [system] Activating service name='org.freedesktop.fwupd

Jan 11 19:12:03 virtualbox-ivan-ubuntu-16

org.freedesktop.fwupd[742]: (fwupd:3801): Fu-WARNING **: Failed to

Jan 11 19:12:03 virtualbox-ivan-ubuntu-16

dbus[742]: [system] Successfully activated service 'org.freedesktop

Jan 11 19:12:12 virtualbox-ivan-ubuntu-16

pulseaudio[3595]: [pulseaudio] bluez5-util.c: GetManagedObjects()

Jan 11 19:12:12 virtualbox-ivan-ubuntu-16

gnome-session[2388]: ** (unity-fallback-mount-helper:3682): WARNING

Jan 11 19:12:12 virtualbox-ivan-ubuntu-16

org.gnome.ScreenSaver[2008]: ** Message: Lost the name, shutting d

Jan 11 19:12:22 virtualbox-ivan-ubuntu-16

org.gnome.zeitgeist.Engine[2008]: Performing VACUUM operation... O

Jan 11 19:12:22 virtualbox-ivan-ubuntu-16

org.gnome.zeitgeist.Engine[2008]: ** (zeitgeist-datahub:4005): WAR

Jan 11 19:12:23 virtualbox-ivan-ubuntu-16

com.canonical.Unity.Scope.Applications[2008]: Error loading package

Jan 11 19:12:23 virtualbox-ivan-ubuntu-16

com.canonical.Unity.Scope.Applications[2008]: (unity-scope-loader:

Jan 11 19:13:05 virtualbox-ivan-ubuntu-16

gnome-session[2388]: /var/lib/dpkg/lock:

Jan 11 19:13:12 virtualbox-ivan-ubuntu-16

dbus[742]: [system] Activating service name='org.debian.apt' (usin

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon: INFO: Initializing daemon

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon [INFO]: Initializing daemon

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

dbus[742]: [system] Successfully activated service 'org.debian.apt

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon.PackageKit: INFO: Initializing PackageKit compat layer

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: /usr/lib/python3/dist-packages/aptdaemon/work

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: from gi.repository import PackageKitGlib as

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon.PackageKit [INFO]: Initial

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon: INFO: UpdateCache() was called

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon [INFO]: UpdateCache() was

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

gnome-session[2388]: (gnome-software:3733): GLib-GObject-CRITICAL

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon.Trans: INFO: Queuing transaction /org/debian/apt/transac

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon.Trans [INFO]: Queuing trans

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon.Worker: INFO: Simulating trans: /org/debian/apt/transact

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon.Worker [INFO]: Simulating

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

AptDaemon.Worker: INFO: Processing transaction /org/debian/apt/tra

Jan 11 19:13:14 virtualbox-ivan-ubuntu-16

org.debian.apt[742]: 19:13:14 AptDaemon.Worker [INFO]: Processing

Non-Repudiation



wasn't me...
cross my heart

A close-up photograph showing two hands, one from the left and one from the right, holding a white rectangular piece of paper. The hands are positioned at the top corners of the paper, with fingers slightly curled. The word "Privacy" is printed in a bold, black, sans-serif font in the lower half of the image. The background is a plain, light-colored surface.

Privacy

Attack Types

- Attacks are instances when a security requirement is violated.

Attack/Threat	Security requirement
Spoofing	Authenticity
Tampering	Integrity
Repudiation	Non-repudiation
Information disclosure	Confidentiality
Denial of service	Availability
Elevation of privilege	Authorization

Security Controls/Countermeasures

	Physical	Logical/ Technical	Administrative/ Policies
Preventive (+deterrence)			
Detective			
Corrective/ Responsive			

Controls

	Physical	Logical/ Technical	Administrative/ Policies
Preventive (+deterrence)	Security guard	Firewall	Employee screening policy
Detective	Security camera	Intrusion detection system	Policy to review transaction logs
Corrective/ Responsive	Fire sprinklers	Backup servers	Law enforcement liaison

Security is Asset Protection

Asset

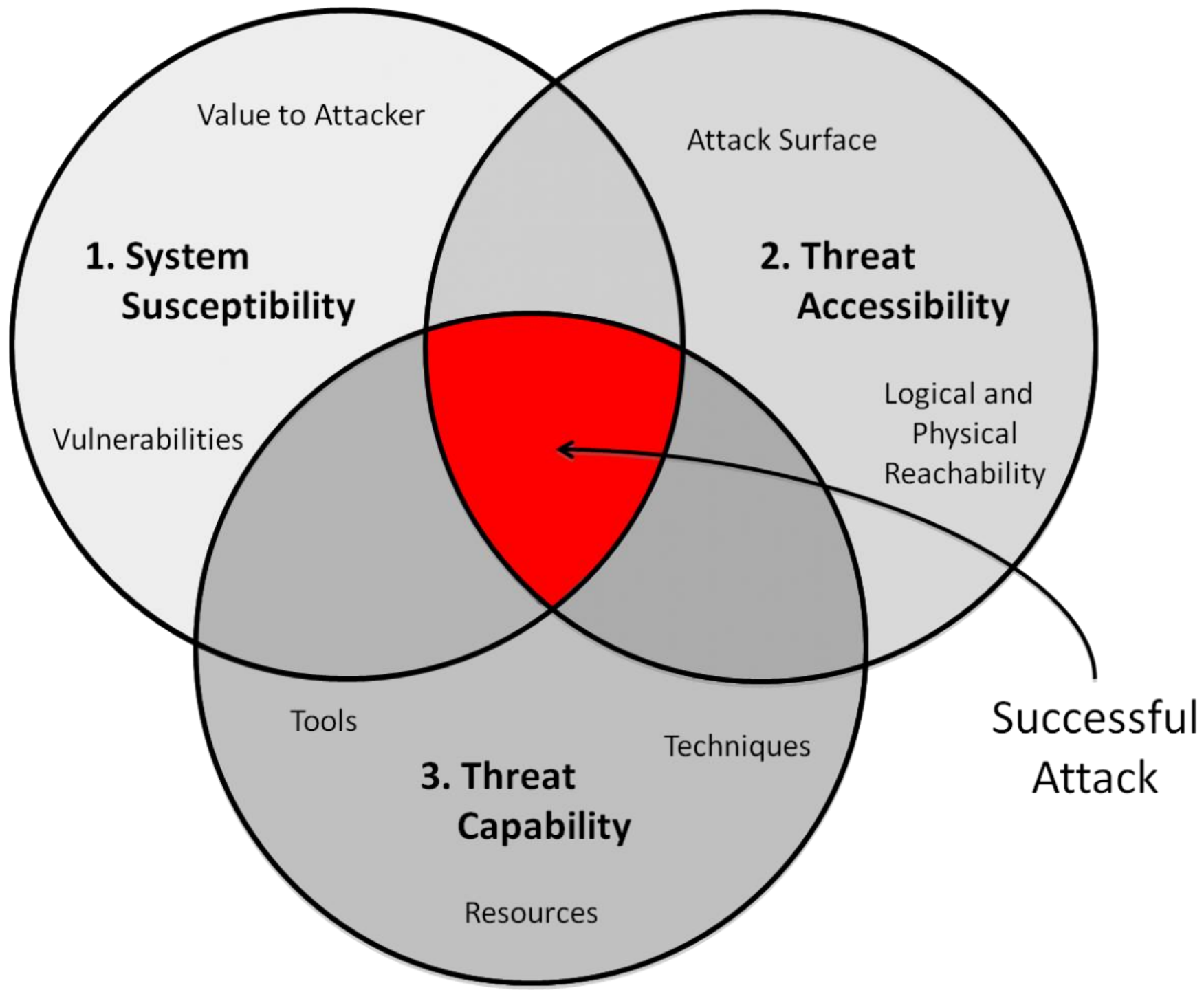


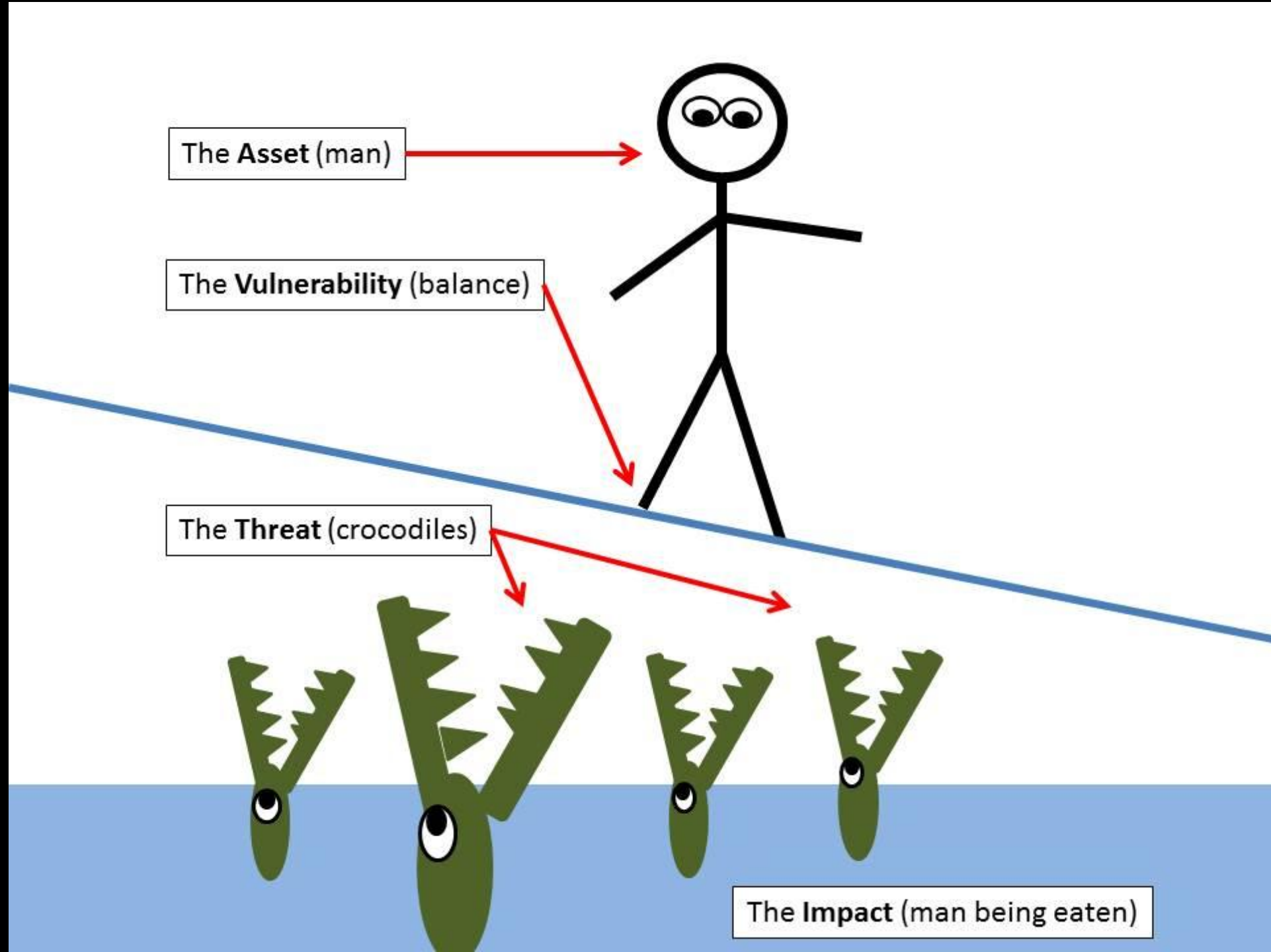
Vulnerability



threat







Trust

Trust This Computer?

Your settings and data will be accessible from this computer when connected wirelessly or using a cable.

Trust

Don't Trust

Security is Hard!

Why is Security hard?

- Consider all potential attacks
- Complex, multi-level, expensive mechanisms
- Security is often an afterthought
- Could be counterintuitive and not user-friendly
- Requires monitoring and capabilities to respond
- Benefits are evident best after a failure

Security is a process, not a target

Security in context

- Security is needed to prevent and counteract the unwanted consequences
- *but security is **not** needed for its own sake*
- *this leads to **trade-offs** or security **dilemmas***

Security Dilemma 1

Security unaware users want security **but** have no security expertise.

Trade-off 2

Security hinders ease of use, **but** if security mechanism is too difficult the users will find a way around it.

Trade-off 3

Security always has a cost, **but** offers no direct gain, only prevention and reduction of loss.

Trade-off 4

Sometimes failure could cost less than prevention.

Security Dilemma 5

Theory might say that an algorithm is secure, **but** its implementation will fail.

and

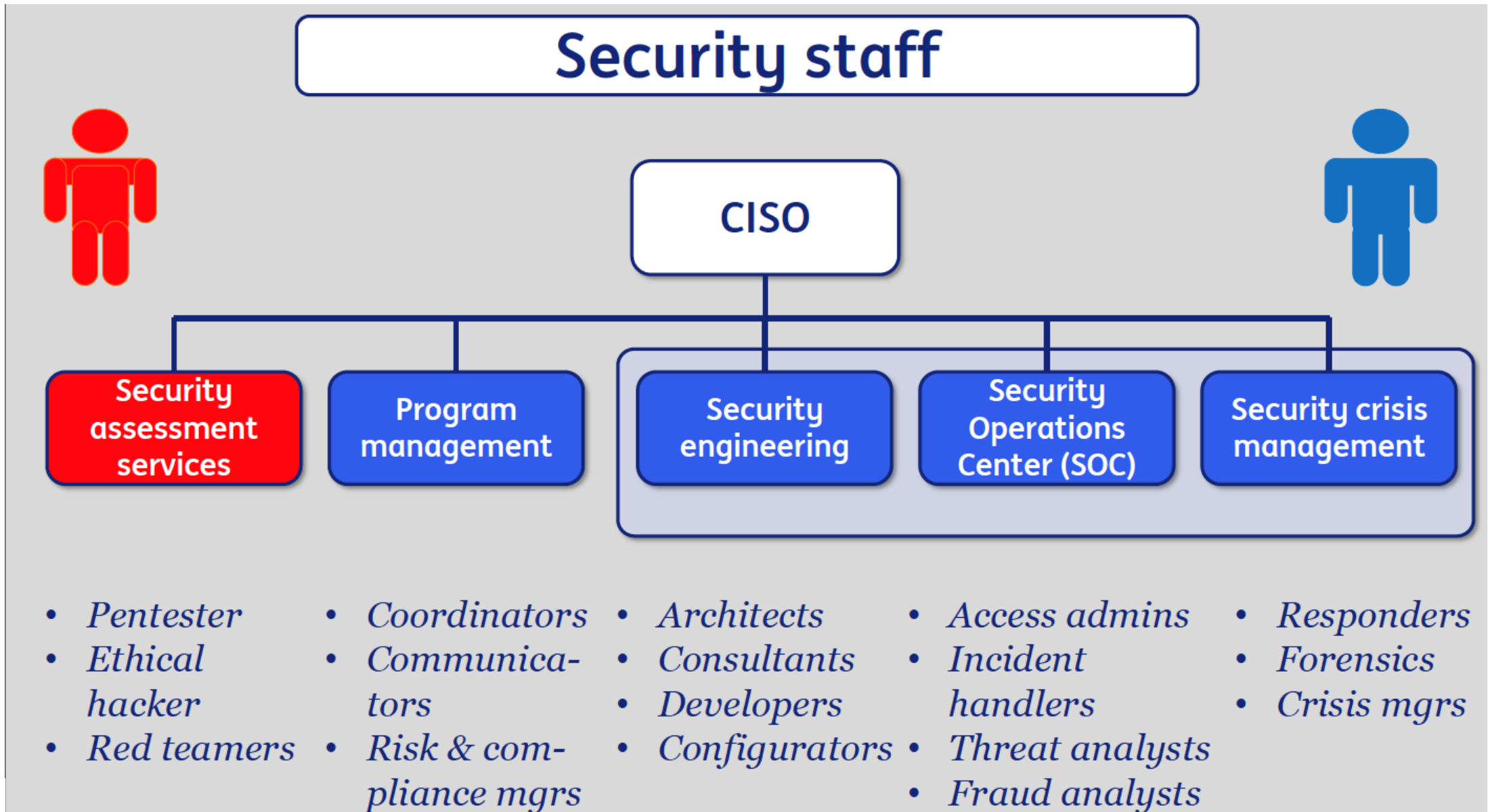
Practical security measures are often imperfect from the theoretical perspective.

Security Dilemma 6

More security increases system complexity, what increases the attack surface.

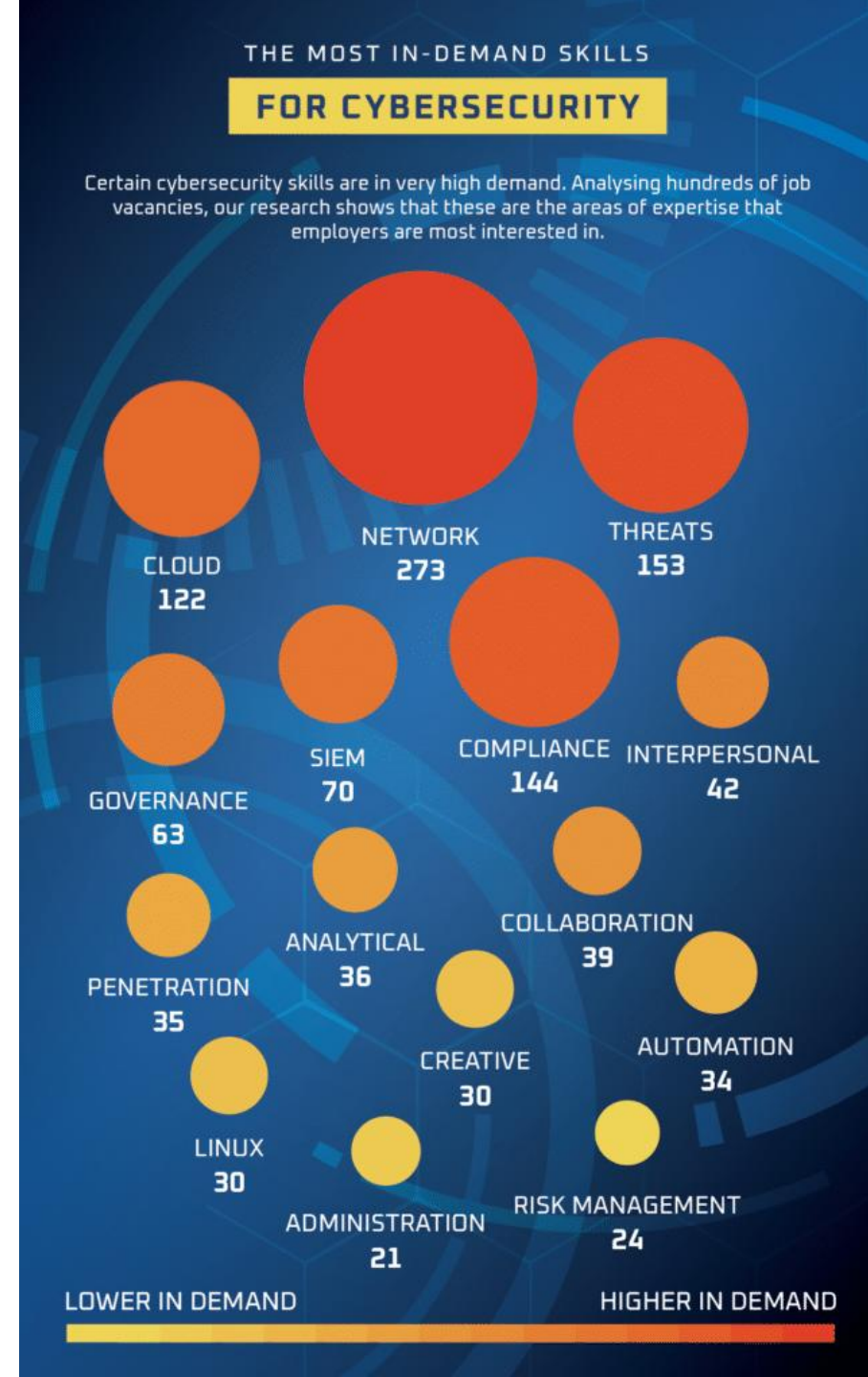
So, what about those security jobs?

Jobs in Security



- We cannot study it all in one course, but we will try to focus on some key elements.
- You will be equipped to continue on any career path in security.

<https://techshielder.com/the-best-cities-for-a-cybersecurity-job>



Course Content

- **Security management:**
 - Threat modelling
 - Security risk management
- **System and application security:**
 - Vulnerabilities
 - Secure software/Application Security
 - Malware analysis
 - Web security
 - Network security
- **Fundamentals:**
 - Access control and identity management
 - Cryptography

Disclaimer:

This is a provisional plan and
may be subject to changes

Questions?



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Bij ons leer je de wereld kennen